ACADEMIC ACTIVITY 2008-2013

Christopher D. Hacon. Distinguished Professor at the Department of Mathematics, University of Utah

Study and research support:

Simons Investigator 2012-2017
Clay Senior Scholar for the MSRI program in Spring 2009;
N.S.F. Grant 2008-2013;

Awards:

2009 Frank Nelson Cole Prize in Algebra
2011 Antonio Feltrinelli Prize in Mathematics, Mechanics, and Applications
2013 Fellow of the AMS

Publications:

[1] *Existence of minimal models for varieties of general type* (With Birkar, Cascini, M^{c}Kernan) J. Amer. Math. Soc. 23 (2010), no. 2, 405-468.

[2] *Existence of minimal models for varieties of general type II: Pl-flips.* Preprint. (With J. M^{c}Kernan) J. Amer. Math. Soc. 23 (2010), no. 2, 469-490.

[3] *On Ueno's Conjecture K.* (With J.A. Chen) Math. Ann. 345 (2009), no. 2, 287296.

[4] *Deformations of canonical pairs and Fano varieties.* J. Reine Angew. Math. 651 (2011), 97126. (With T. deFernex) arXiv:0901.0389

[5] *Singularities on normal varieties.* (With T. deFernex) Compos. Math. 145 (2009), no. 2, 393–414.

[6] *On the geography of threefolds of general type.* (With J.A. Chen) Jour. of Algebra **321** (2009) 2500-2507.

[7] *The Sarkisov Program.* To appear in JAG. (With J. M^{c}Kernan)

[8] *The canonical ring is finitely generated.* Colloquium De Giorgi 2007 and 2008, Umberto Zannier editor, Edizioni della Scuola Normale Superiore di Pisa.

[9] *Higher dimensional minimal model program for varieties of general type.* Analytic and algebraic geometry, 525555, IAS/Park City Math. Ser., 17, Amer. Math. Soc., Providence, RI, 2010.

[10] *Factoring 3-fold flips and divisorial contractions to curves.* (With J.A. Chen). Journ. Reine Angew. Math. 657 (2011), 173197.

[11] *Rigidity properties of Fano varieties.* (With T. deFernex) Current developments in algebraic geometry, 113127, Math. Sci. Res. Inst. Publ., 59, Cambridge Univ. Press, Cambridge, 2012. arXiv:0911.0504

[12] *Classification of higher dimensional algebraic varieties.* (With S. Kovács). Oberwolfach seminars, 41. Birkhuser Verlag, Basel, 2010. x+208 pp..

[13] *On the birational automorphisms of varieties of general type.* (With J. M^c^Kernan and C. Xu) arXiv:1011.1464, to appear in Annals.

[14] *Boundedness of varieties of general type.* (With J. M^c^Kernan and C. Xu). In preparation.

[15] *Boundedness results in birational geometry* (With J. M^c^Kernan). ICM 2010 proceedings.

[16] *Flips and flops.* (With J. M^c^Kernan) ICM 2010 proceedings.

[17] *Kodaira dimension of irregular varieties.* (With J. A. Chen) Invent. Math. 186 (2011), no. 3, 481–500.

[18] *Extension theorems, Non-vanishing and the existence of good minimal models.* (With J.-P. Demailly and M. Paun) To appear in ACTA arXiv:1012.0493

[19] *The acc for log canonical thresholds.* (With J. M^c^Kernan and C. Xu). arXiv:1208.4150.

[20] *Existence of log canonical closures.* (With C. Xu.) Inventiones Mathematicae 2012; arXiv:1105.1169.

[21] *On Finiteness of B-representation and Semi-log Canonical Abundance.* (With C. Xu.) arXiv:1107.4149.

[22] *On log canonical inversion of adjunction.* To appear in PEMS, volume in honour of V. Shokurov. arXiv:1202.0491

[23] *Non-rational centers of log canonical singularities.* To appear in Jour. of Algebra. arXiv:1109.4164 (With V. Alexeev.)

[24] *Singularities of pluri-theta divisors in Char $p > 0$.* To appear in Alg. Geom. East Asia. arXiv:112.2219

[25] *On the numerical dimension of pseudo-effective divisors in positive characteristic.* (with P. Cascini, M. Mustata and K. Schwede. arXiv:1206.4150)

[26] *Generic vanishing fails for singular varieties in characteristic $p > 0$.* (With J. Kovacs. arXiv:1212:5105)

[27] *On the three dimensional minimal model program in positive characteristic.* (With C. Xu. arXiv:1302.0298)

Algebraic Geometry Seminars: Univ. of Georgia Athens 2008, Univ. of Texas at Austin 2008, IMPA Brasil 2009.

Colloquiums: Univ. of Georgia Athens 2008, Rice Univ. 2008, Univ. of Texas at Austin 2008, UVU 2010, Berkley-Stanford A.G. 2010, UW 2011, La Sapienza Rome 2011; UW-PIMS Mathematical Colloquium 2012.

Distinguished lectures: Phillips Lecture Series MSU 2008, Opening Symposium of the Global Center of Excellence Tokyo Jan. 2009, MSRI Evans Lecture April 2009, Plenary Talk AMS Fall 2009 Meeting UCR, Plenary Talk British Mathematical Colloquium in 2010 at Edinburgh, Zassenhaus Lecture Series at Ohio State 2010, Lecture in the Algebraic and complex geometry session of the 2010 ICM, Frontiers of Science Univ. of Utah 2011, Plenary lecture 2012 ECM Krakow.

Conferences: 2008 Texas Algebraic Geometry Seminar, 2008 Park City Summer School (5 talks), 2008 Oberwolfach seminars (8 lectures), 2009 MSRI Classical Algebraic Geometry, 2010 sectional talk at the British Mathematical Colloquium in 2010 at Edinburgh, Pacific Rim (at Stanford) 2010, AGNES (at MIT) 2011, AIM 2012, 2012 ECM satellite conference Krakow, 2013 Kawamata's 60th birthday conference in Tokyo.

Other professional activities and service:

Editor for the Journal of Algebraic Geometry.
Associate Editor for the Journal of the American Math Society.
Associate Editor for the Annals of Mathematics.
MSRI Science Advisory Committee 2012-2016;
Chern Distinguished lecture committee 2011-present.
Refereed for many international journals.
Reviewed papers for Zentralblatt Math.
Reviewed grant proposals for NSF, NSA, PRIN (Italy).
Wrote letters of recommendation for students, post doc, tenure track and tenured faculty etc.

Organized the following conferences: 1) Oberwolfach seminars 2008, 2) Algebraic Geometry: session of the AMS meeting, Riverside, 2010, 3) Algebraic Geometry: session of the AMS meeting, Salt Lake City 2011; 4) AIM workshop "The ACC for MLD's and the termination of flips", 5) WAGS Fall 2012, 6) Robfest 2013 (Robert Lazarsfeld's 60-th conference), 7) Oberwolfach Algebraic Geometry conference 2013.

Served on several departmental committees including: Executive committee 09-10, 10-11; Hiring committee 11-12; Undergraduate Awards and Scholarships Committee 08-12; Hiring committee chair 11-12; Ad Hoc RPT Committee for T. de Fernex 08; Ad Hoc RPT Committee for K. Wortman 09; Ad Hoc RPT Committee for Y.P. Lee 10; Ad Hoc RPT Committee for A. K. Singh 10; Review committee for C. Xu 2012;

Ph.D. Students and their papers:

S. Urbinati (2012 Ph.D.; Post. Doc. in Poland): *"Discrepancies of non-$\mathbb{Q}$-Gorenstein varieties"* Mich. Jour. Math. *"Divisorial models of normal varieties. "* arXiv:1211.1692

C.-J. Lai (2012 Ph.D.; Post. Doc at Purdue): *"Varieties fibered by good minimal models"* Math. Ann.; *"Bounding the volumes of singular Fano threefolds"* Preprint.

X. Jiang: (2014 Ph.D - expected) *" On the pluricanonical maps of varieties of intermediate Kodaira dimension.* arXiv:1012.3817

O. Das.: (2015 Ph.D - expected)

Y. Zhang. (2014 Ph.D - expected) *"Pluri-canonical map of varieties of maximal Albanese dimension in positive characteristic."* arXiv:1203.1360; *"On the Volume of Isolated Singularities."* arXiv:1301.6121

Y. Wang (2015 Ph.D - expected)

A. Watson (2015 Ph.D - expected)

Courses Taught:

2008-2009:	Math 1260	AP calculus I
2009-2010:	Math 1040	Introduction to probability and statistics (2 sections)
2010-2011:	Math 5310-5320	Introduction to Modern Algebra
2011-2012:	Math 1040	Introduction to probability and statistics (2 sections)
2012-2013:	Math 2210	Calculus III
	Math 2270	Linear Algebra
	Math 7800	Topics in Algebraic Geometry